

4.4 A 2.1-to-3.7ppm/°C Bandgap Voltage Reference with a Current-Domain TC Compensation and $\pm 0.06\%$ Inaccuracy from -40°C to 125°C in 130nm CMOS

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Abstract

This paper presents a CMOS bandgap reference (BGR) with a current-domain high-order TC compensation by using a capacitively-biased-diode-based super-PTAT current bias. It achieves a TC of 2.1 to 7.1ppm/°C from -40 to 125°C after a batch trim, which is on par

with state-of-the-art BGRs with a 1-point trim. After a 2-point trim, it achieves a TC that ranges from 2.1 to 3.7ppm/°C and an inaccuracy of $\pm 0.06\%$ (3σ) from -40 to 125°C. Its simple architecture also results in 2× higher energy efficiency.

BJT-based bandgap references (BGRs) are essential building blocks for measurement and metering chips. Their 1st-order temperature coefficient (TC) can be minimized by adding a PTAT voltage (ΔV_{BE}) to a CTAT voltage (V_{BE}) in the correct proportions. However, due to the nonlinear term ($\propto T \cdot \ln(T)$) in V_{BE} , high-order TCs remain, which limit the accuracy of BGRs in standard CMOS processes [1-9]. To avoid this, a nonlinear compensation voltage V_{NL} can be added to the BGR output (Fig. 4.4.1 top). This can be generated by a nonlinear PTAT voltage ΔV_{BE} [1-3], piecewise-linear compensation [4,5], and other nonlinear voltage-generation circuits [6,7]. However, such methods usually result in larger noise and spread, which limit the energy efficiency and accuracy of the BGR. Although better accuracy can be achieved with a digital look-up-table (LUT) and a temperature sensor [8], the compensated result is only available in the digital domain. As mentioned in [9], the curvature in V_{BE} can also be suppressed by using a strongly temperature-dependent bias current ($\propto T^n$). However, generating such a super-PTAT current requires either translinear circuits based on high-quality BJTs [10], or V_{PTAT}/R current-generation circuits based on a resistor with a large negative TC ($\sim -0.33\%/^{\circ}\text{C}$) [11], which are not available in most standard CMOS processes.

In this work, a CMOS BGR with a current-domain high-order TC compensation is presented. The curvature of V_{BE} is compensated by a tunable super-PTAT bias current through a capacitively-biased-diode (CBD)-based current generator. Implemented in a standard 130nm CMOS process, the proposed BGR achieves a TC of 2.1 to 7.1ppm/°C from -40 to 125°C after a simple batch trim, which is on par with state-of-the-art precision BGRs with a 1-point trim [3]. After a 2-point trim, it achieves a TC that ranges from 2.1 to 3.7ppm/°C and an inaccuracy of $\pm 0.06\%$ (3σ) from -40 to 125°C. Compared to state-of-the-art precision BGRs [2-6], its simple architecture results in 2× higher energy efficiency.

Figure 4.4.1 (bottom) shows the diagram of the proposed current-mode BGR with a current-domain TC compensation. The reference output V_{REF} is generated by adding an amplified PTAT voltage ΔV_{BE} to a CTAT voltage V_{BE} . Apart from the classic PTAT bias current I_{PTAT} , a super-PTAT bias current I_{BIAS} is also injected into the emitter of the BJT Q_3 , compensating for the curvature of V_{BE} directly. Compared with other voltage-domain TC compensation schemes [2-7], in which the voltage error is linearly added to the BGR, this current-domain TC compensation is less sensitive to errors, as any variation in I_{BIAS} will only lead to a logarithmic change in V_{BE} . The main challenge is then how to generate such a super-PTAT bias current in a simple way.

To avoid the use of BJT-based translinear circuits, a CBD-based super-PTAT bias generation approach is proposed that only requires a few transistors and capacitors (Fig. 4.4.1 bottom-right). As in [12], the capacitor C ($\approx 6.2\text{pF}$) is first pre-charged to V_{DD} during the reset (Φ_{RST}) phase. Then it is discharged via a diode-connected PMOS M_D . After a short settling time (tens of ns), the current through M_D is proportional to $kT/q \cdot C/t$. Since the on-chip MIM capacitor C has a negligible TC, the current is PTAT when a fixed discharging time t is used [13-15]. However, if t decreases with temperature, the temperature dependency of the current will be significantly increased, thus achieving super-PTAT behavior. Moreover, due to the diode characteristic, this current is also insensitive to the initial voltage on the capacitor C , resulting in a good power-supply rejection (PSR) [12]. After discharging, the current of M_D is sampled and replicated via the current-mirror transistor M_1 to bias the BJT. By tuning the 1st-order TC of the discharging time t , the TC of the super-PTAT I_{BIAS} can be tuned to compensate for the curvature of V_{BE} . In this approach, the TC compensation is finally accomplished in the time-domain, which simplifies the analog design.

As in [16], the discharging time t can be defined by an on-chip RC delay. To compensate the 2nd-order TC in V_{BE} , t should have a large negative TC ($\sim -0.35\%/^{\circ}\text{C}$) [11]. Unfortunately, no standard CMOS resistor can provide this temperature characteristic. Instead, this is achieved by using logic gates to subtract a positive TC delay t_2 from a smaller negative TC delay t_1 ($t_1 > t_2$, Fig. 4.4.2 top). Figure 4.4.2 (bottom) shows the diagram of the tunable negative-TC delay-generation circuit. It consists of two modified RC polyphase filters (PF₁ and PF₂) and two inverter-based comparators (COMP₁ and COMP₂) [16]. The polyphase filters are implemented with unsilicided N-poly resistors ($R_N = 46.8\text{k}\Omega$, $TC = -0.1\%/^{\circ}\text{C}$), silicided N-poly resistors ($R_P = 14.8\text{k}\Omega$, $TC = -0.28\%/^{\circ}\text{C}$) and MIM capacitors C_1 and C_3 ($\approx 3.1\text{pF}$). As in [16], the extra capacitors C_2 and C_4 ($= C_1 = C_3$) prevent the occurrence of beyond-the-rail output spikes, thus allowing the comparators to be realized with core

devices without over-voltage stress issues. In this work, the auxiliary comparator COMP₁ with a relaxed speed is used to detect the zero-crossing (ZC) moment of PF₂ ($V_{AP} = V_{AN}$). Its output (SEL) is then used to select the input of the main comparator COMP₂. Before the ZC moment of PF₂, the input of the COMP₂ is connected to PF₂. After that, it is switched to PF₁. The time interval t between the ZC moments of the two PFs can then be extracted from the output of COMP₂. Since COMP₂ is reused for both PFs, its switching delay and offset are cancelled, making the time interval t purely defined by the RC time constants. To save power in the inverter-based comparator, reset switches are used to pull V_{AP}/V_{AN} and V_{BP}/V_{BN} up/down to V_{DD} /ground, respectively, shortly after the comparator switches [16]. In case the high-order TCs of the BJT are not well-modeled, the TC of the discharging time t and thus the BGR's curvature is adjustable by a 4-bit tunable R_N .

The generated negative TC delay is used to control the CBD-based super-PTAT bias generator. However, the CBD bias cannot provide a stable output during the reset and discharging phases (Φ_{RST} and Φ_{DC}). To solve this issue, a ping-pong technique [12] is adopted to provide a continuous-time bias current for the PNP Q_3 in the BGR core (Fig. 4.4.3). To mitigate the impact of the leakage current, the bias current is refreshed at $f = 200\text{kHz}$. With a discharging time of 165ns, the simulated super-PTAT I_{BIAS} is about 9μA at room temperature and increases to 16μA at 125°C. For simplicity, I_{BIAS} is also mirrored to bias the error amplifier A in the BGR core. Thanks to the simple structure of the CBD-based bias generator, it can easily start up as long as a clock is available, thus avoiding additional start-up circuits, which are usually necessary for conventional BGRs. The PNP Q_1 and Q_2 have an emitter ratio of 15 (unit size $5\mu\text{m} \times 5\mu\text{m}$), which defines the PTAT bias current ($I_0 = \Delta V_{BE}/R_1$). It is further mirrored to R_2 and Q_3 , thus generating the curvature-compensated output $V_{REF} = 2R_2/R_1 \cdot \Delta V_{BE} + V_{BE}$. To ensure that the emitter current of Q_3 is dominated by the super PTAT I_{BIAS} , the normal PTAT current ($2I_0 = 3.6\mu\text{A}$) is much smaller than I_{BIAS} . R_2 is 6-bit tunable, so that the PTAT voltage can be tuned to compensate for the 1st-order TC variation. To reduce the offset and 1/f noise of the error amplifier, it is chopped at 200kHz. In addition, the current mirrors are cascoded to improve the PSR. Their mismatch is mitigated by using dynamic element matching (DEM).

Implemented in a standard 130nm CMOS process, the BGR occupies an active area of 0.1mm^2 (Fig. 4.4.7). It generates a 1.87V output V_{REF} with a 3.3V analog supply and a 1.2V digital supply for the delay-generation circuit. To mitigate measurement errors and suppress the chopping and DEM ripples, the output is followed by an on-chip chopped buffer and then filtered by an external 1μF capacitor. Figure 4.4.4 (top-left) shows the measured power consumption of the BGR over temperatures. At room temperature, it consumes 91μA and 10μA in the analog and digital domains, respectively. Due to the super-PTAT bias current, the analog current increases to 177μA at 125°C. Figure 4.4.4 (top-right) shows the measured output noise of the BGR. In the frequency band of interest (0.1Hz to 10Hz), the noise can be significantly suppressed by chopping and DEM, resulting in an integrated noise of 1μVrms. Changing the analog supply from 2.4V to 3.6V only changes the BGR output by 200μV, corresponding to a power-supply sensitivity (PSS) of less than $0.016\%/V$ (8 samples, Figure 4.4.4 bottom-left). Varying the digital supply from 1.15V to 1.25V causes $\pm 0.06\%$ error in V_{REF} due to the residual supply dependency of the RC delay, which can be mitigated by the LDO. By injecting a 100mVp AC signal to the analog supply, it shows a PSR of -90dB at 10Hz.

To verify the high-order TC compensation of the proposed BGR, a typical sample is first characterized from -40 to 125°C. As shown in Fig. 4.4.4 (bottom-right), the residual high-order TC of the BGR can be tuned by adjusting R_N in the RC delay-generation circuit. With the optimal trimming code, it achieves a residual TC of 2.6ppm/°C. The same high-order TC trimming code is applied to all the samples. 16 samples from one batch with DIL ceramic package were characterized over temperature. Without any individual trimming, the BGR has an untrimmed inaccuracy of $\pm 0.26\%$ (3σ) from -40 to 125°C (Fig. 4.4.5, top-left). The TCs of different samples vary from 2.1ppm/°C to 7.1ppm/°C with an averaged value of 4.6ppm/°C (Fig. 4.4.5, top-right). The inaccuracy is reduced to $\pm 0.13\%$ (3σ) after a 1-point offset trim at room temperature. To further improve the accuracy, 2-point calibration at 20°C and 95°C can be used to trim the 1st-order TC by tuning R_2 , the residual TC is then reduced to 2.1 to 3.7ppm/°C with an averaged value of 2.7ppm/°C (Fig. 4.4.5, bottom). The BGR then achieves an inaccuracy of $\pm 0.06\%$ (3σ) from -40 to 125°C. To investigate the

effect of packaging stress, 16 samples from the same batch were characterized in SOP plastic packaging, resulting in a small shift in the optimal trimming code. Thanks to the added polyimide layer in the die, it achieves a similar inaccuracy of $\pm 0.25\%$ (3σ) and a TC of 2.9 to 8.1 ppm/ $^{\circ}\text{C}$ with a batch trim. After 2-point calibration for the 1st-order TC and nominal value trim, the inaccuracy can be further improved to $\pm 0.05\%$ (3σ) from -40 to 125 $^{\circ}\text{C}$.

Figure 4.4.6 summarizes the performance of this work and compares it to state-of-the-art precision BGRs in standard CMOS processes. Thanks to the direct current-domain TC compensation scheme, this work achieves the lowest untrimmed TC, which is on par with the state-of-the-art precision BGR with a 1-point trim [3]. It also achieves low noise of 1 μVrms from 0.1Hz to 10Hz while consuming 101 μA , improving the energy efficiency by 2 $\times$ due to its simple structure.

Acknowledgement:

This work was supported by the National Natural Science Foundation of China under Grant 62522409. The corresponding author is Zhangming Zhu (zmyh@263.net).

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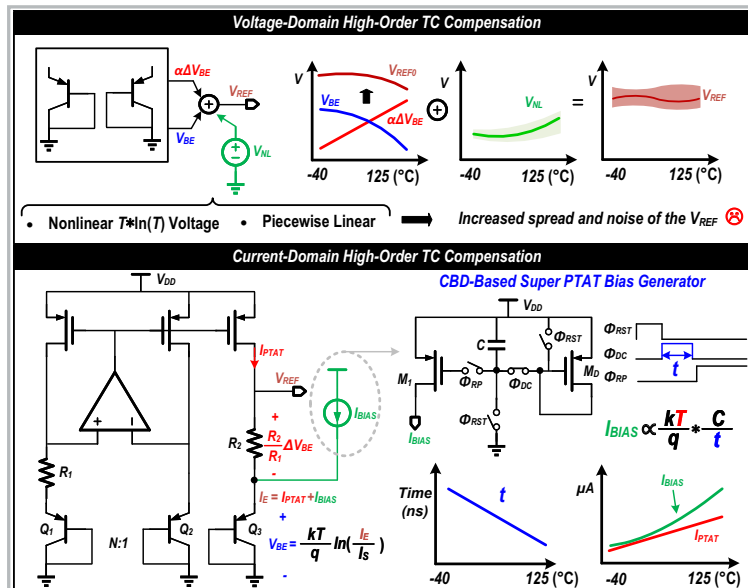


Figure 4.4.1: Conventional BGR with an added non-linear compensation voltage V_{NL} (top); the proposed BGR with a current-domain high-order TC compensation (bottom).

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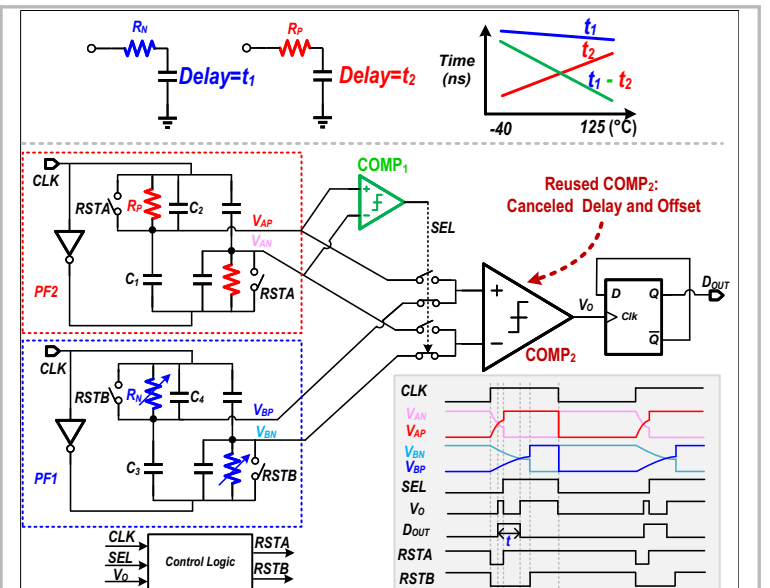


Figure 4.4.2: Concept of the on-chip negative TC delay generation (top) and its circuit implementation (bottom).

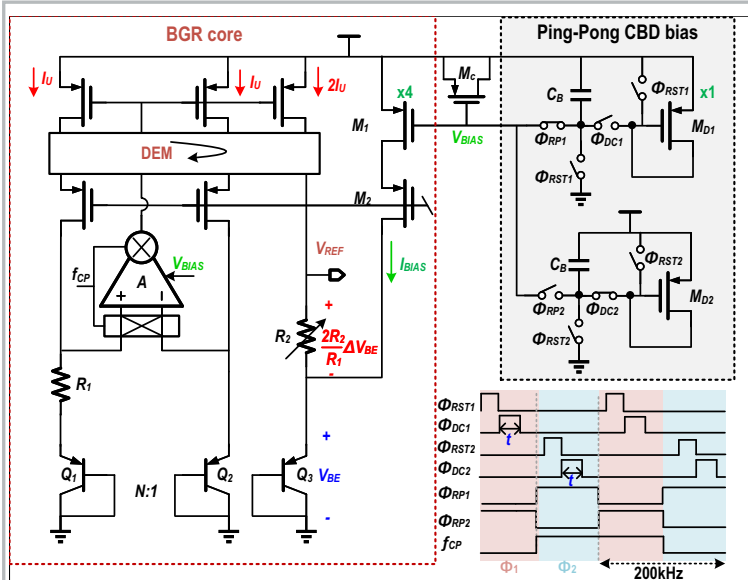


Figure 4.4.3: Circuit and timing diagrams of the current-mode BGR with a ping-pong super-PTAT bias.

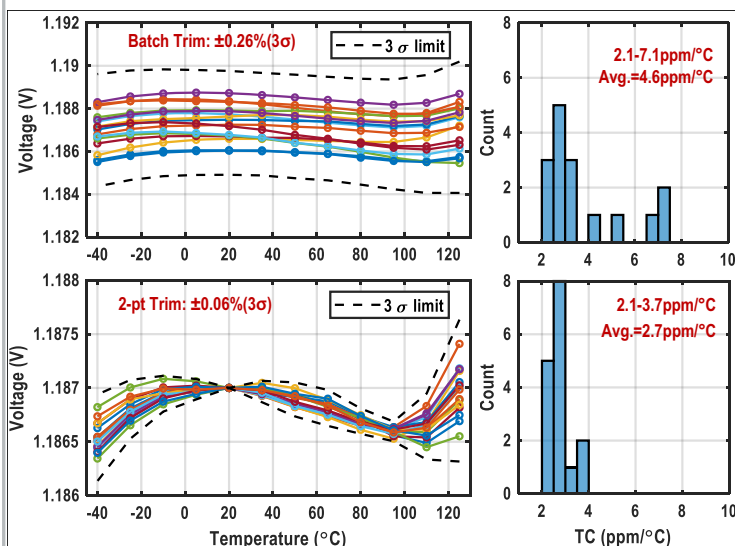


Figure 4.4.5: Measured inaccuracy of 16 samples in ceramic package after a batch trim (top-left) and their residual TCs (top-right); inaccuracy after a 2-point trim (bottom-left) and their residual TCs (bottom-right).

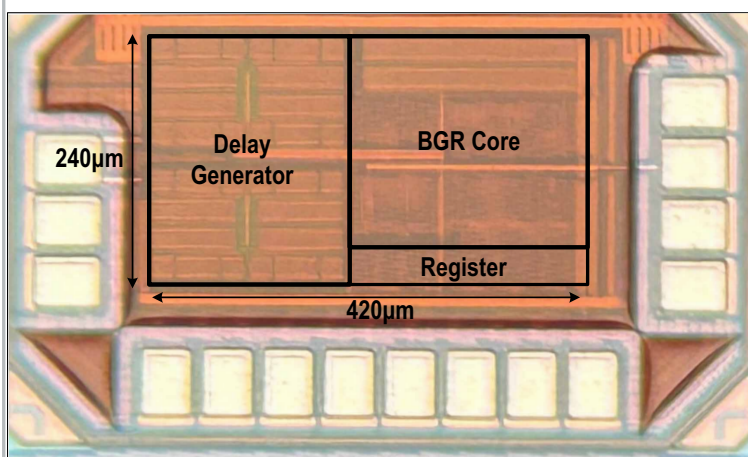


Figure 4.4.7: Die micrograph.

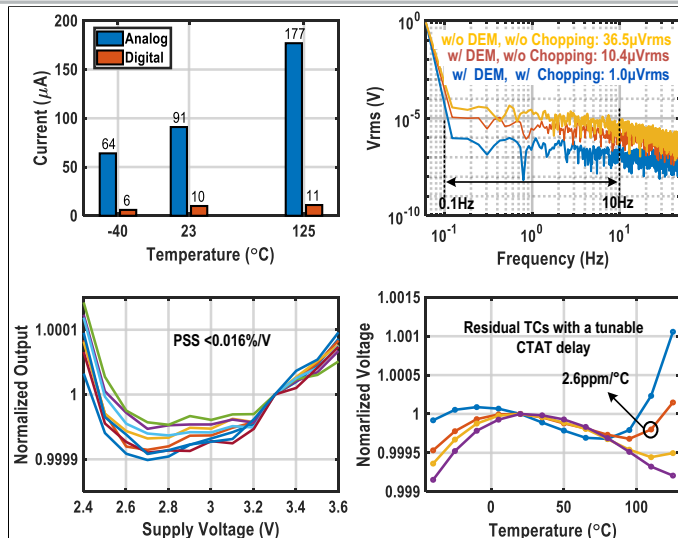


Figure 4.4.4: Measured current consumption at different temperatures (top-left); measured output noise with different DEM and chopping settings (top-right); measured power supply sensitivity (PSS) of 8 samples (bottom-left); measured residual TCs with different CTAT delay times of one typical sample (bottom-right).

	This work	JSSC'11 [2]	JSSC'21 [3]	TCAS-I'22 [6]	JSSC'21 [4]	ESSCIRC'23 [5]	ISSCC'15 [8]
Technology (nm)	130	160	180	180	130	40	130
Area (mm ²)	0.10	0.12	0.38	0.256	0.08	0.18	0.034
Curvature Correction	Super PTAT Current	Non-Linear Voltage	Non-Linear Voltage	Non-Linear Voltage	Piecewise Linear	Piecewise Linear	Digital LUT
Supply (V)	2.4-3.6/1.2	1.8±10%	1.8±10%	3.2-3.7	3.3	1.4-1.8	1.5
Bandgap Voltage (V)	1.187	1.0875	1.14	2.14	1.16	1.2	1.215
Temp. range (°C)	-40 to 125	-40 to 125	-40 to 125	-25 to 125	-40 to 150	-40 to 125	-40 to 120
3σ Inaccuracy (Trim Point)	±0.26% (0) ±0.13% (1) ±0.06% (2)	±0.75% (0) ±0.15% (1)	±0.21% (0) -0.02%/ +0.12% (1)	N/A	±1.62% (0)	±0.35% (0)* ±0.1% (1)*	±0.08% (1)
TC (ppm/°C) (Trim Point)	2.1-7.1 (0) 2.1-3.7 (2)	5-12 (1)	3.2-5.5 (1)	7.4-20.3 (0) 0.7-1.53 (3)	5.78-13.5 (0)	6.5-9.6 (1)	7 (1)
Current (μA)	91/10	55	17	409	120	56	N/A
Noise (μV) @BW	1@10Hz	6.1@10Hz	56@2.5Hz	15@1kHz	175.5@10Hz	1.9@10Hz	N/A
Line Sensitivity (%/V)	0.016	N/A	N/A	0.0146	0.03	0.19	N/A
PSR (dB)	-90@10Hz	-74@DC	-76dB@1Hz	-63.4@10Hz	-82@10Hz	-77@10Hz	N/A
Efficiency (μA×μV ² /Hz)**	10.1	204.7	21324.8	92.0	369603	20.2	N/A
Samples	16 Ceramic + 16 Plastic	61	18	6	7	10	13 Ceramic

*Estimated from figures; **Efficiency=Current × Noise²/Hz

Figure 4.4.6: Performance summary and comparison with the state-of-the-art.